

Assignment (2)

Evaluate the following integrals:

$$1. \int \frac{2}{1+x^2} dx = 2 \int \frac{1}{1+x^2} dx = 2 \tan^{-1}(x) + C$$

$$2. \int \frac{dx}{x^2 + 4x + 4} = \int \frac{1}{(x+2)^2} dx = \frac{(x+2)^{-1}}{-1 \times 1} + C$$

$$\left(x + \frac{4}{2} \right)^2 + 4 - \left(\frac{4}{2} \right)^2 = (x+2)^2$$

$$3. \int \frac{dx}{x\sqrt{x^2-4}} = \int \frac{1}{2x\sqrt{\frac{x^2}{4}-1}} dx = \frac{1}{2} \int \frac{1}{x\sqrt{\left(\frac{x}{2}\right)^2-1}} dx = \frac{1}{2} \sec^{-1}\left(\frac{x}{2}\right) + C$$

$$4. \int \frac{dx}{(x-1)\sqrt{x^2-2x+2}} = \frac{1}{(x-1)^2+1}$$

$$= \frac{1}{(x-1)\sqrt{(x-1)^2+1}} \Rightarrow \frac{1}{1+(x-1)^2}$$

$$= \tan^{-1}(x-1)$$

$$5. \int \frac{dx}{x^2+4x+8} = \frac{1}{(x+2)^2+4} = \int \frac{1}{4\left(1+\left(\frac{x+2}{2}\right)^2\right)} dx$$

$$(x+2)^2+4$$

$$\frac{1}{4} \int \frac{1}{1+\left(\frac{x+2}{2}\right)^2} dx \rightarrow \frac{1}{4} \int \frac{1}{1+\left(\frac{x+2}{2}\right)^2} dx$$

$$\frac{1}{4} \int \frac{1}{1+\left(\frac{x+2}{2}\right)^2} dx = \frac{1}{4} \tan^{-1}\left(\frac{x+2}{2}\right) + C$$

$$6. \int \frac{dx}{\sqrt{9+4x-x^2}} = -x^2+4x+9 \Rightarrow -[x^2-4x-9]$$

$$(x^2-4x-9) \Rightarrow (x-2)^2-13$$

$$= \int \frac{1}{\sqrt{(x-2)^2-13}} \Rightarrow \frac{1}{\sqrt{13}} \int \frac{1}{\sqrt{\left(\frac{x-2}{\sqrt{13}}\right)^2-1}}$$

$$= \frac{1}{\sqrt{13}} \cosh^{-1} \left(\frac{x-2}{\sqrt{13}} \right) \Rightarrow \frac{1}{13} \frac{\cosh^{-1} \left(\frac{x-2}{\sqrt{13}} \right)^2}{\frac{1}{\sqrt{13}}} + C$$

$$7. \int (\sin 2x + \cos 2x)^2 dx = \int \sin^2 2x + \cos^2 2x + 2 \sin 2x \cos 2x dx$$

1 Sin 4x

$$\int 1 + \sin 4x dx$$

$$= x - \frac{\cos 4x}{4} + C$$

$$8. \int \sin 3x \cos 4x \, dx = \frac{1}{2} \int \sin(-x) + \sin(7x) \, dx$$

$$= \frac{1}{2} \left[\frac{-\cos(-x)}{-1} - \frac{\cos(7x)}{7} \right] + C$$

$$9. \int \operatorname{sech} x (\cosh^3 x + \tanh x) \, dx$$

$$= \int \cosh^2 x + \operatorname{sech} x \tanh x \, dx$$

$$= \int (\cosh x \cosh x) + \operatorname{sech} x \tanh x \, dx$$

$$= \int \frac{1}{2} \left[\cosh(2x) + \underbrace{\cosh(0)}_1 \right] + \operatorname{sech} x \tanh x \, dx$$

$$\frac{1}{2} \left[\frac{\sinh(2x)}{2} + x \right] - \operatorname{sech} x + C$$

$$10. \int (2 + \tan x)^2 dx \quad \int 4 + 4 \tan x + \tan^2 x dx$$

$$4 \int \tan x = \frac{\sin x}{\cos x} = -\ln(\cos x) = 4 \ln(\sec x) + C$$

$$\int 4 dx = 4x$$

$$\tan^2 x + 1 = \sec^2 x \rightarrow \sec^2 x - 1 = \tan^2 x$$

$$\int \tan^2 x = \int (\sec^2 x - 1) dx = \int \sec^2 x = \tan x$$

$$= \int -1 = -x$$

$$= 4x + [4 \ln(\sec x)] + \tan x - x + C$$

$$11. \int \frac{\cos^2 x}{1 + \sin x} dx$$

$$\cos^2 x = (1 - \sin^2 x) \rightarrow \int (1 - \sin x)(1 + \sin x)$$

$$= \int \frac{(1 - \sin x)(1 + \sin x)}{1 + \sin x} dx = \int 1 - \sin x dx$$

$$= x + \cos x + C$$

$$12. \int \cosh 2x \cosh 5x \, dx = \frac{1}{2} \int [\cosh(7x) + \cosh(-3x)] \, dx$$

$$= \frac{1}{2} \left[\frac{\sinh(7x)}{7} - \frac{\sinh(-3x)}{3} \right] + C$$

$$13. \int \frac{-\operatorname{csch}^2 y + \cosh y}{\coth y + \sinh y} \, dy = \ln(\coth y + \sinh y) + C$$